Identification and characterization of merger debris through clustering using Gaia data

A project to get you started in Galactic archaeology

Identification and characterization of merger debris through clustering using Gaia data

Where do we expect to find merger debris?

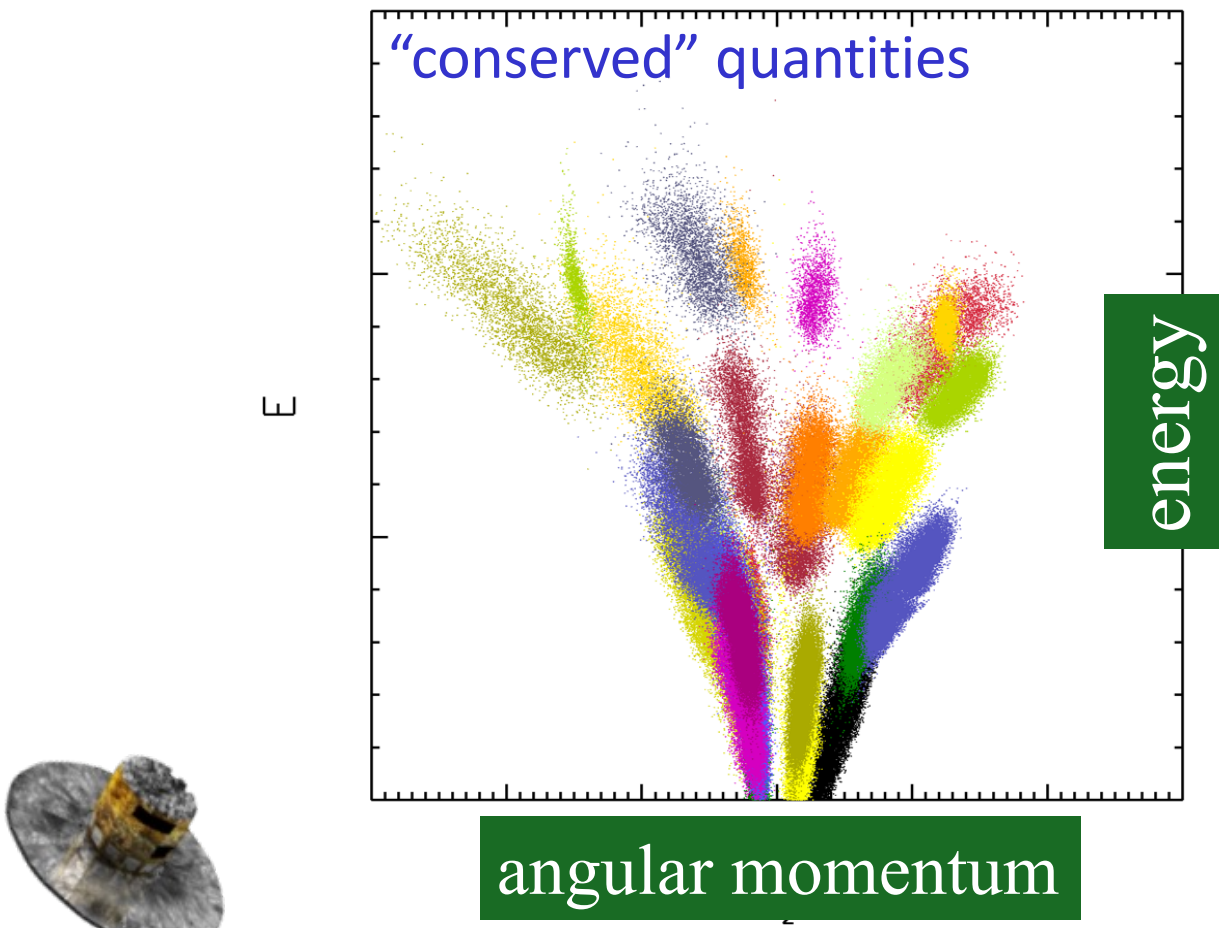
Stellar halo

How can we construct a halo sample?

Kinematically using Gaia data

How do we find merger debris?

How do we find merger debris?



What do we need?

- Compute Energy (MW potential), Lz and Lperp* from observables (proper motions, parallaxes, line-of-sight velocities)
- Use a clustering algorithm
- Assess significance of clusters

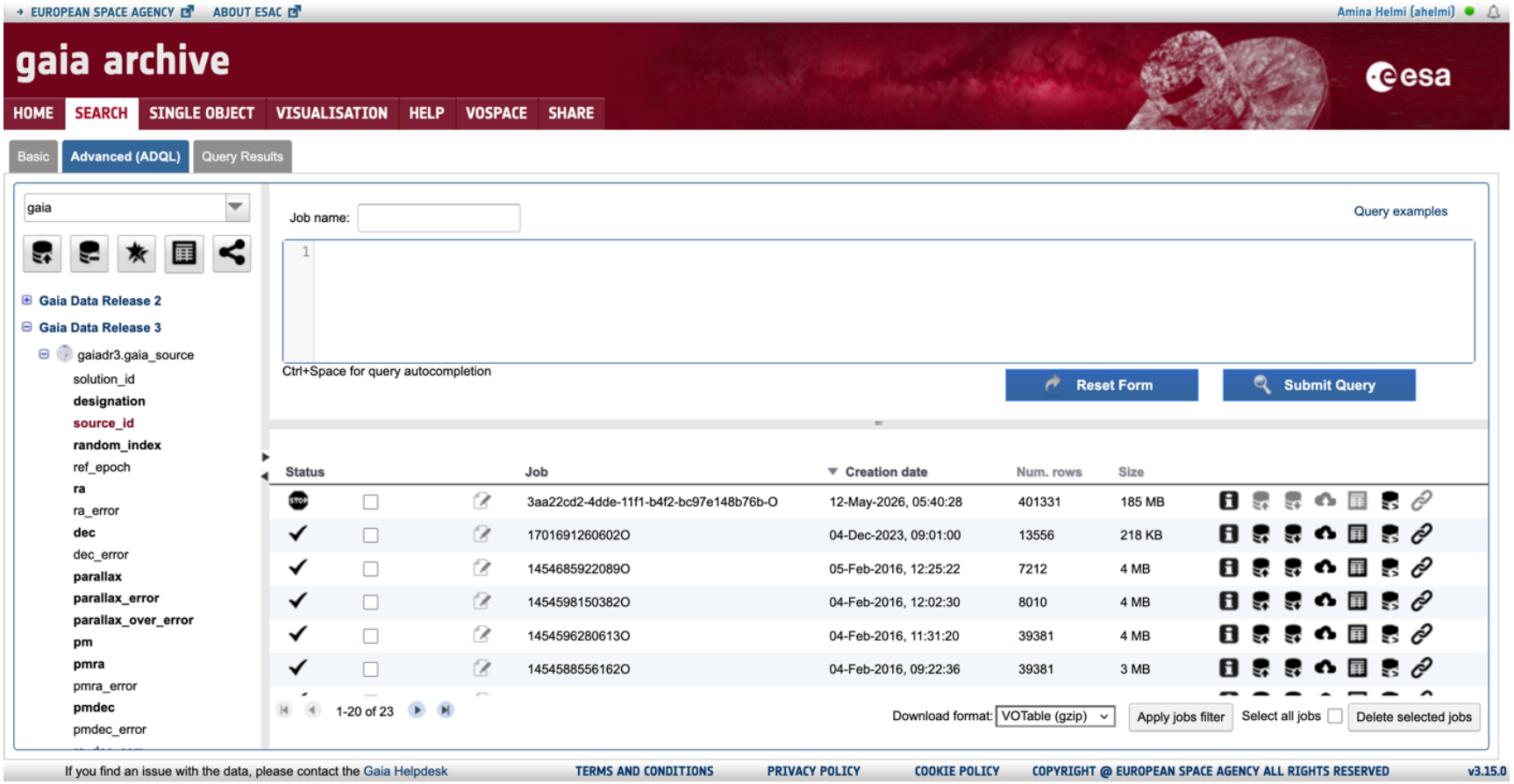
* Lperp = $(Lx^2 + Ly^2)^{1/2}$

Constructing a sample of nearby stars

- The Gaia archive provides many different measurements (with uncertainties and parameters)

- We can use:
- DATALABS
- Gaia archive
- Filtered file

Either approach is fine,
(quickest the file provided)
Remember
DR4 is Dec 2nd !!



Constructing a sample of nearby stars

- What we need is 6D phase-space coordinates and magnitudes G, and GBP , GRP information for nearby stars
- the 6D phase-space (x,y,z,vx,vy,vz) are inferred from (ra,dec,distance) and ($\boldsymbol{\mu}_{ra}$*, $\boldsymbol{\mu}_{dec}$, v_{los})

- IMPORTANT:
  - Only a subset of the stars has v_{los} measured `(radial_velocity IS NOT NULL)`
  - Gaia does not measure distance but parallax. We will use stars with parallax/err_{par} > 5 (a 20% parallax error) → distance ~ 1/parallax

- Quality cuts/corrections:
  i. parallax corrected by individual zero-point offset (Lindegren et al. 2021).
  ii. Distance = 1/ $\boldsymbol{\varpi}_{corr}$ for stars with $(\varpi - \Delta_\varpi)/\sqrt{\sigma^2_{parallax} + \sigma^2_{sys}} \geq 5,$

e.g. selecting stars with measured v_{los}

```
SELECT source_id, ra, dec, parallax, pmra, pmdec,
    radial_velocity, radial_velocity_error, grvs_mag, ....
FROM gaiadr3.gaia_source
WHERE radial_velocity IS NOT NULL
```

```
from astroquery.gaia import Gaia


query = """

SELECT source_id, ra, dec, radial_velocity,
radial_velocity_error, ...

FROM gaiadr3.gaia_source

WHERE radial_velocity IS NOT NULL

"""

job = Gaia.launch_job_async(query)   # use login()
first for large results

tbl = job.get_results()
```

Constructing a sample of nearby stars

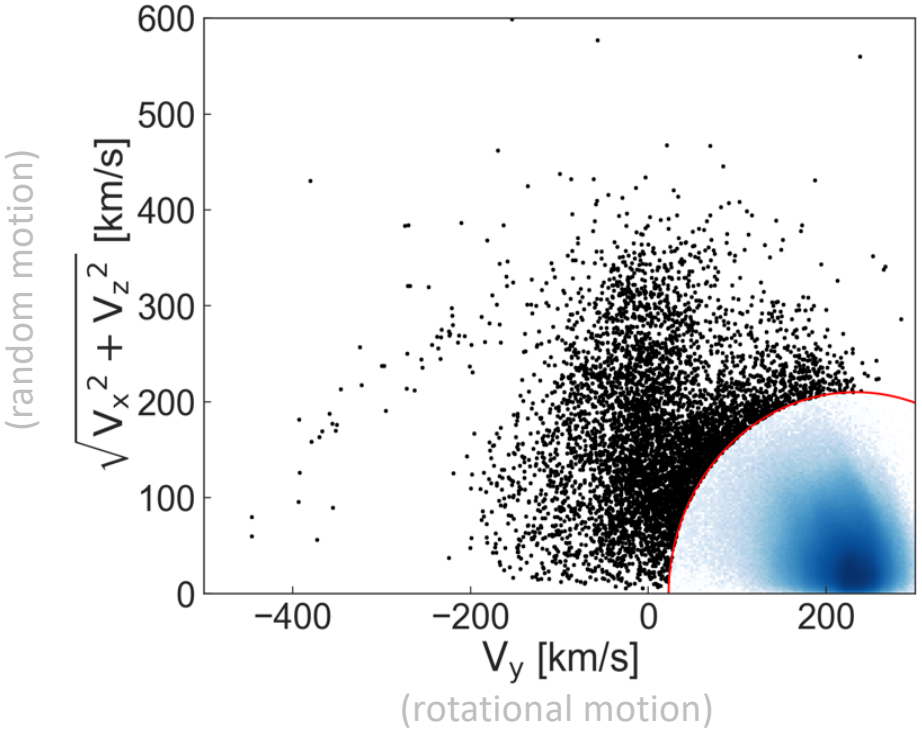
- Quality cuts/corrections (cont.)
  iii. RUWE < 1.4 and $\boldsymbol{\sigma}(v_{los}) < 20$ km/s, after applying correction to radial_velocity_error (Babusiaux et al. 2023).
  iv. removed a few stars with $(G_{RVS} - G) < 3$.
- The file provided has all the stars (satisfying i and ii) but not iii) and iv)

- Now select stars within 2.5 kpc from the Sun
- Compute velocities (transformation from observables to heliocentric velocities and correct for LSR and peculiar motion of the Sun)
- Compute Galactocentric positions (`astropy`)

- Next construct a halo sample

Constructing a halo sample *near the Sun*

Toomre diagram



A **kinematic** selection

$v_x^2 + (v_y - v_{LSR})^2 + v_z^2 > V^2_{thresh}$

- Value of V_{thresh} depends on location (typically ~ V_{circ}) → V_{thresh} = 180 km/s
- Biased against debris on low-inclination, disk-like orbits

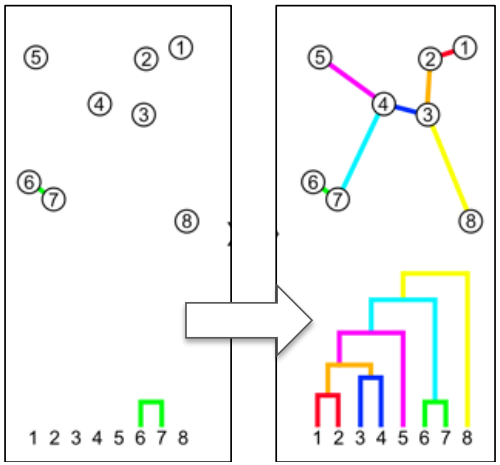
Compute Energy (requires assuming Milky Way potential), L_z and L_{perp}

Identification of substructures through a data-driven clustering algorithm

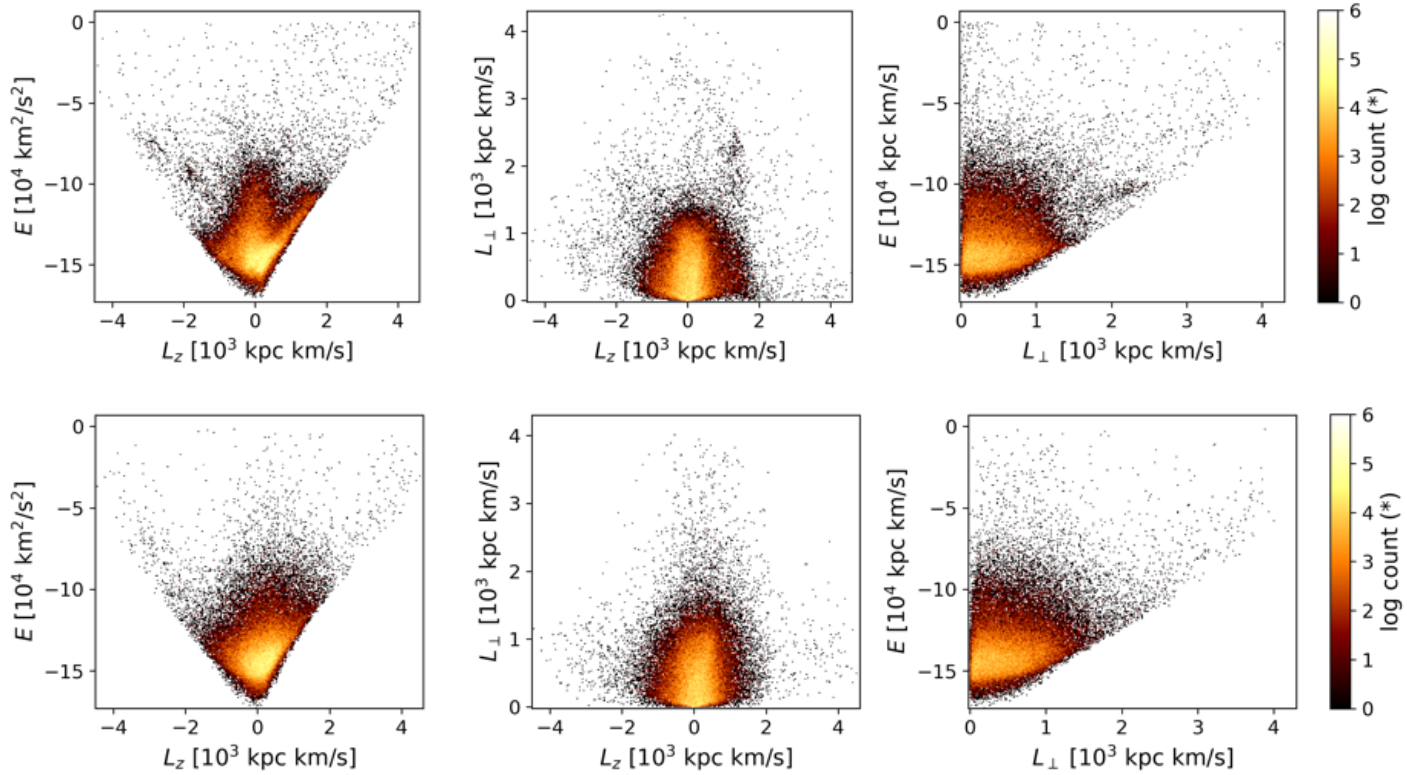
Single-linkage algorithm
applied to 3D space
defined by Integrals of Motion

Clusters are built hierarchically
using distances between stars

- every star is in a cluster



RANDOM SET

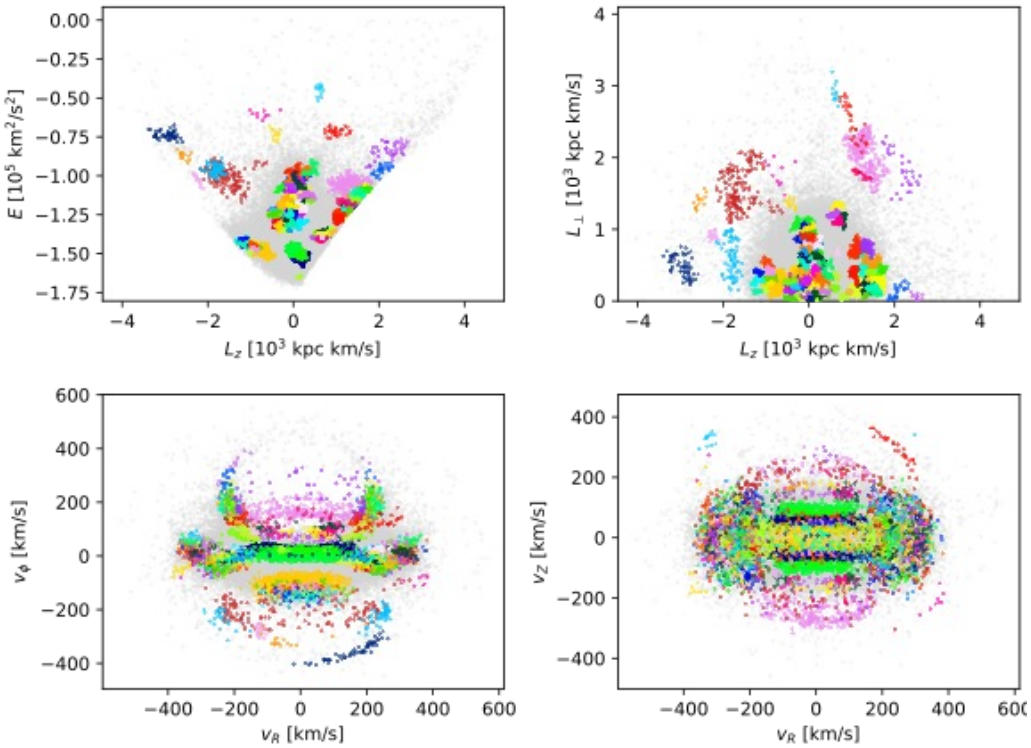


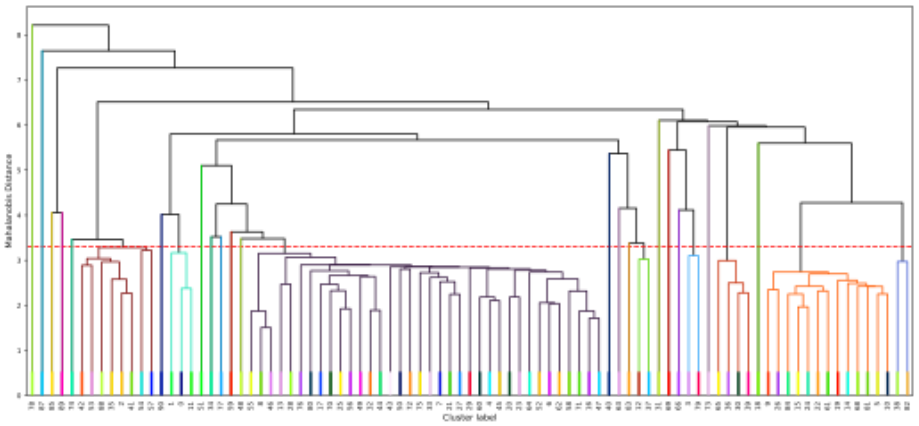
Compare extent in 3D to randomized datasets to determine significance
(randomization by reshuffling the velocities)

Lövdal et al 2022, Ruiz-Lara et al (2022); Dodd et al (2023)

Identification of substructure
(Gaia DR3)







- 91 clusters found
- 20% of the dataset

Important amount of fragmentation → group clusters into larger structures

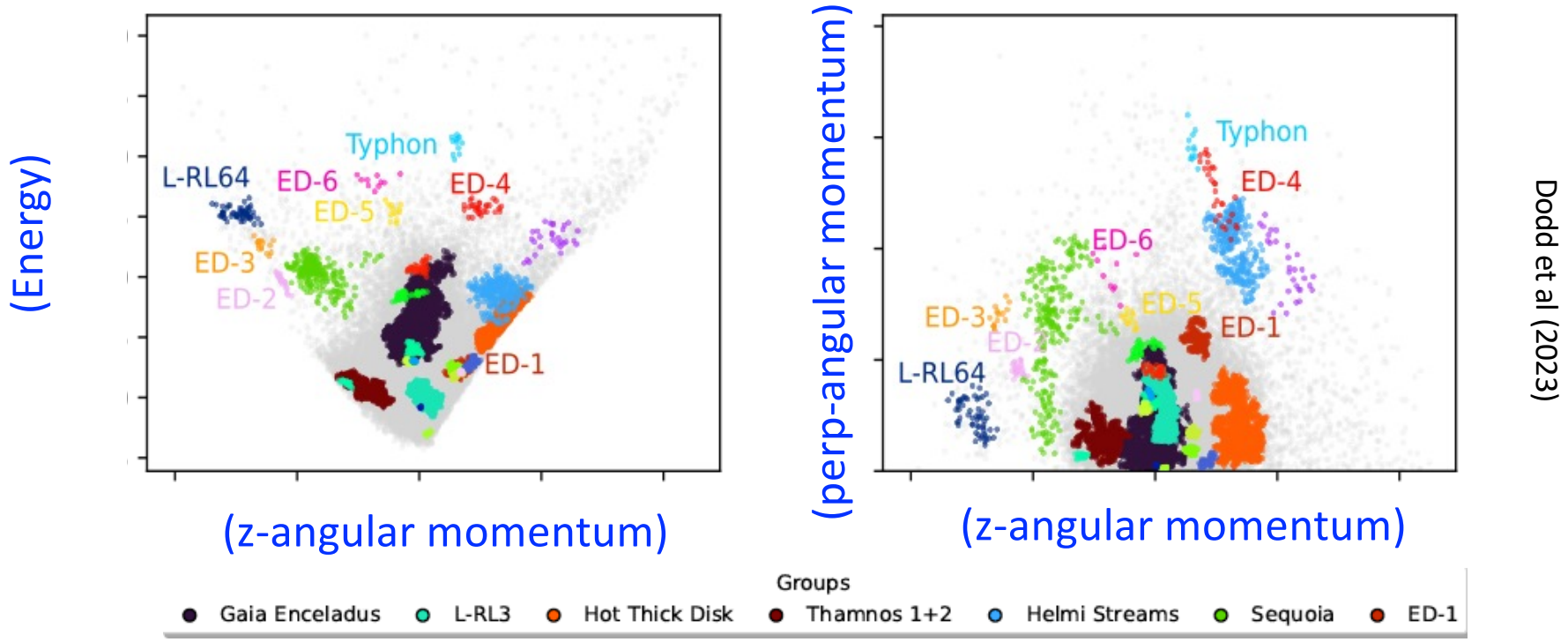
Lövdal et al 2022, Ruiz-Lara et al (2022); Dodd et al (2023)

Large substructures and small clusters



Emma Dodd

(Potentially) the stars from the systems that built the early Milky Way



Dodd et al (2023)

→ **True "archaeology"**

also Myeoung et al; Naidu et al.; Horta et al. Yuan et al.

Stars IDs and characteristics of the structures publicly available

Resources

DATA

- Gaia archive: https://gea.esac.esa.int/archive/
- Datalabs https://datalabs.esa.int
- Dataset gaia-dr3-rv-poege5.hdf5, (34Gb). The data can be downloaded from

https://filesender.surf.nl/?s=download&token=2fde27c7-3bf2-48f5-ab34-ae7bb3498ea2

SOFTWARE

- https://github.com/SofieLovdal/IOM_clustering
- https://github.com/TomCallingham/id$_$Dodd23/blob/main/README.md

Follow steps from Dodd et al. (2023)
https://scixplorer.org/abs/2023A%26A...670L...2D/abstract

Extensions of this project

- Extend the analysis to beyond the Solar neighbourhood
  - requires revisiting selection of halo stars, see e.g. Posti et al., 2018, Veljanoski et al., 2019
- Complement Gaia data with spectroscopic surveys for v_{los} (and [Fe/H])
  - DESI, APOGEE, LAMOST, GALAH [see, e.g. Ruiz-Lara et al., 2022, Koposov et al., 2026].
- Search for additional members using Gaia data without v_{los}
  - Integrate orbits forwards and backwards in time of known members of given substructure
  - Characterize nature of substructure using CMD or metallicity distribution function [see e.g. Lahoye, 2025, Nogueira-Santos et al., 2026].
  - Characterize Mass function [Offner, 2026, Li et al., 2026]
- Use Gaia DR4 when released

With big thanks to them!



500+ astronomers/engineers

